

Developing Data-Driven Models for Real-Time Dredging Production Optimization

1. Job Description Field in Smart Recruiters

Boskalis is currently looking for a student to start their graduation project with us. Dredging vessels rely on trained crews to configure dredging process settings that directly affect production output, a task that involves numerous interacting physical processes and a wide range of operating conditions. We currently use a machine learning model to recommend operation settings to crews in real time, but there is clear room for improvement. Physics-informed machine learning, time-series modelling, and adaptive control methods are promising directions worth exploring, as they can capture relationships and patterns that the current approach does not yet take advantage of.

The research project will require the student to conduct a literature review across these fields and determine which methods best fit our data and problem. Key design choices need to be made, such as how to combine physical knowledge with data-driven models, and how to validate improvements using historical data. With a variety of techniques available, an educated choice must be made to effectively improve our existing system.

In this research you will

- Work with other data scientists and meet with multi-disciplinary engineers to understand the challenge.
- Conduct literature research on physics-informed machine learning, time-series modelling, and adaptive control, and how they can be applied to this problem.
- Formulate the problem in a way that combines physical knowledge with data-driven models.
- Develop methods to improve the model's recommendations and demonstrate how they generalize across different vessels and conditions.
- Report your final findings, demonstrating your design choices and reproducibility of results.

Your qualities

- You are doing a university master's degree in computer science, AI, mathematics, physics, or a related field.
- Affinity with Python and version control.
- Able to develop, train, and evaluate ML models.
- Theoretical understanding of physics-informed ML, time-series modelling, or adaptive control; practical knowledge is not required but preferred.
- Proficiency in writing reports and findings, and presenting results.
- It is preferred to work at our office in Papendrecht minimally 3 days a week.

2. Qualifications Field

The resources that you will be able to use

- Receive supervision and support from AI and data science experts from the AI department.
- Gain access to necessary data and compute resources for training and evaluating models (Databricks).
- Low threshold to plan meetings with people to gain a better understanding of the problem.
- Utilize well-established development facilities (Python, source code control, packaging).
- Access and review other developments that the AI department is working on.

What you can expect

- Graduation/internship guidance: We offer you the opportunity to get the most out of your internship by giving you the right guidance.
- Warm welcome: You can count on a warm welcome so that you quickly feel at home at Boskalis.
- As an intern, you will receive an internship allowance. In addition, we offer a fun work environment with lots of challenges.

- A dynamic work environment: An internship where you can learn a lot from a leading company and where you will be part of a diverse team of experts.

Young Boskalis: Are you younger than 36? Then join Young Boskalis! Monthly social and sporting activities ranging from pub quizzes, yoga, bootcamps and an annual sailboat race. Networking and knowledge sharing are also an important part of Young Boskalis.

3. Company Description

About Boskalis

Working at Boskalis is about creating new horizons and sustainable solutions. In a world where population growth, increase of global trade, demand for (new) energy and climate change are driving forces, we challenge you to make your mark in finding innovative and relevant solutions for complex infrastructural and marine projects.

Within a vibrant company culture, you will be working as part of a diverse, international team of experts. We offer you the opportunity to realize your full personal potential and expand your professional career by creating new horizons. Together.

4. Additional Information Field in Smart Recruiters

Commencement: Open to discussion

Do you have questions about this graduation project or Interested?

Then send an email to campus@boskalis.com
Or contact amine.berjaoui.tahmaz@boskalis.com